COLUMN DESIGN SUMMARY

Project Name	:	G+1 RB COLUMN DESIGN
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Column-2.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 03:00:45

Note:

- Some Of the checks For vertical And horizontal reinforcement are updated In latest version Of RCDC, Thus If the RCDC file created Using older version (older than V11 Update 1), Is opened In latest version And calculation reports are directly generated, Then, some Of the values printed In the report may Not match the design. For such files created Using older version, it Is recommended To unlock the old design And re-perform the same

Design Code: IS 456 : 2000 + IS 13920 : 2016

Load Combinations :

L/C No	Analysis No	Load Definition
1	5	1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)
2	6	1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL)
3	7	1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)
4	8	1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)
5	9	1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) -1.2 (LOAD 1: EQ X)
6	10	1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) -1.2 (LOAD 2: EQ Z)
7	11	1.5 (LOAD 3: DL)
8	12	1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
9	13	1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)
10	14	1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
11	15	1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)
12	16	0.9 (LOAD 3: DL)
13	17	0.9 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)
14	18	0.9 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)
15	19	0.9 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)
16	20	0.9 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)

Levels :

1. FOUNDATION
2. 1.981 m
3. 5.029 m
4. 8.077 m
5. 11.125 m
6. 13.625 m
7. 14.125 m

Column/Wall : **C1**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
1 TO 2	230 X 450	M25 : Fe550 : Fe550	13	17	36.79	3.3	36	1.17	0.76	6-T16	T8 @ 225
2 TO 3	230 X 450	M25 : Fe550 : Fe550	13	17	24.25	-3.29	41.65	1.17	0.9	6-T16	T8 @ 225
3 TO 4	230 X 450	M25 : Fe550 : Fe550	15	19	95.43	11.78	39.74	1.17	0.79	6-T16	T8 @ 225
4 TO 5	230 X 450	M25 : Fe550 : Fe550	13	17	5.6	0.45	-26.09	1.17	0.58	6-T16	T8 @ 225

Column/Wall : **C2**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
1 TO 2	230 X 450	M25 : Fe550 : Fe550	8	12	281.85	2.27	48.38	1.99	0.63	4-T20 + 4-T16	T8 @ 225
2 TO 3	230 X 450	M25 : Fe550 : Fe550	8	12	188.96	-2.98	67.41	1.99	0.86	4-T20 + 4-T16	T8 @ 225
3 TO 4	230 X 450	M25 : Fe550 : Fe550	8	12	96.83	6.53	-58.91	1.99	0.8	4-T20 + 4-T16	T8 @ 225
4 TO 5	230 X 450	M25 : Fe550 : Fe550	8	12	2.02	-8.52	-52.03	1.99	0.75	4-T20 + 4-T16	T8 @ 225

Column/Wall : **C3**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
1 TO 2	230 X 450	M25 : Fe550 : Fe550	8	12	558.79	-5.2	37.46	1.17	0.69	6-T16	T8 @ 225
2 TO 3	230 X 450	M25 : Fe550 : Fe550	15	19	125.02	-4.07	-43.36	1.17	0.83	6-T16	T8 @ 225
3 TO 4	230 X 450	M25 : Fe550 : Fe550	15	19	97.62	6.44	37.3	1.17	0.74	6-T16	T8 @ 225

4 TO 5	230 X 450	M25 : Fe550 : Fe550	10	14	115.66	5.21	37.3	1.17	0.72	6-T16	T8 @ 225
6 TO 7	230 X 450	M25 : Fe550 : Fe550	8	12	4.05	-10.39	18.49	1.17	0.37	6-T16	T8 @ 225

Column/Wall : **C4**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
1 TO 2	230 X 450	M25 : Fe550 : Fe550	13	17	102.8	8.69	40.62	1.17	0.81	6-T16	T8 @ 225
2 TO 3	230 X 450	M25 : Fe550 : Fe550	13	17	70.52	9.35	48.15	1.17	0.98	6-T16	T8 @ 225
3 TO 4	230 X 450	M25 : Fe550 : Fe550	13	17	44.8	-8.01	-41.85	1.17	0.87	6-T16	T8 @ 225
4 TO 5	230 X 450	M25 : Fe550 : Fe550	8	12	27.48	-16.88	-33.15	1.17	0.66	6-T16	T8 @ 225

Column/Wall : **C5**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
1 TO 2	230 X 450	M25 : Fe550 : Fe550	8	12	222.15	6.94	53.22	1.99	0.71	4-T20 + 4-T16	T8 @ 225
2 TO 3	230 X 450	M25 : Fe550 : Fe550	8	12	143.22	7.97	74.16	1.99	0.99	4-T20 + 4-T16	T8 @ 225
3 TO 4	230 X 450	M25 : Fe550 : Fe550	8	12	62.14	8.83	65.06	1.99	0.91	4-T20 + 4-T16	T8 @ 225
4 TO 5	230 X 450	M25 : Fe550 : Fe550	8	12	-23.89	-16.42	-54.05	1.99	0.8	4-T20 + 4-T16	T8 @ 225

Column/Wall : **C6**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links

1 TO 2	230 X 450	M25 : Fe550 : Fe550	10	14	491.59	11.93	-42.79	1.6	0.67	4-T20 + 2-T16	T8 @ 225
2 TO 3	230 X 450	M25 : Fe550 : Fe550	10	14	364.95	13.26	-57.5	1.6	0.89	4-T20 + 2-T16	T8 @ 225
3 TO 4	230 X 450	M25 : Fe550 : Fe550	10	14	245.62	-11.35	51.3	1.6	0.8	4-T20 + 2-T16	T8 @ 225
4 TO 5	230 X 450	M25 : Fe550 : Fe550	10	14	119.41	-21.85	48.87	1.6	0.76	4-T20 + 2-T16	T8 @ 225
6 TO 7	230 X 450	M25 : Fe550 : Fe550	8	12	8.15	12.54	17.32	1.6	0.26	4-T20 + 2-T16	T8 @ 225

Column/Wall : **C7**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
1 TO 2	230 X 450	M25 : Fe550 : Fe550	13	17	72.37	9.47	42.77	1.6	0.71	4-T20 + 2-T16	T8 @ 225
2 TO 3	230 X 450	M25 : Fe550 : Fe550	13	17	48.32	11.44	50.93	1.6	0.85	4-T20 + 2-T16	T8 @ 225
3 TO 4	230 X 450	M25 : Fe550 : Fe550	13	17	31.09	-8.45	-43.95	1.6	0.74	4-T20 + 2-T16	T8 @ 225
4 TO 5	230 X 450	M25 : Fe550 : Fe550	9	13	18.96	-63.23	-4.41	1.6	0.46	4-T20 + 2-T16	T8 @ 225

Column/Wall : **C8**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
1 TO 2	230 X 530	M25 : Fe550 : Fe550	8	12	212.1	10.64	62.93	2.39	0.65	8-T20 + 2-T16	T8 @ 225
2 TO 3	230 X 530	M25 : Fe550 : Fe550	8	12	137.33	11.9	85.68	2.39	0.88	8-T20 + 2-T16	T8 @ 225
3 TO 4	230 X 530	M25 : Fe550 : Fe550	8	12	60.76	8.94	73.82	2.39	0.77	8-T20 + 2-T16	T8 @ 225
		M25 :									

4 TO 5	230 X 530	Fe550 : Fe550	8	12	-20.13	-26.67	-51.33	2.39	0.51	8-T20 + 2-T16	T8 @ 225

Column/Wall : **C9**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
1 TO 2	230 X 450	M25 : Fe550 : Fe550	15	19	157.71	9.81	-42.96	1.6	0.69	4-T20 + 2-T16	T8 @ 225
2 TO 3	230 X 450	M25 : Fe550 : Fe550	10	14	236.62	17.11	-57.61	1.6	0.9	4-T20 + 2-T16	T8 @ 225
3 TO 4	230 X 450	M25 : Fe550 : Fe550	10	14	135.6	-14.26	50.85	1.6	0.82	4-T20 + 2-T16	T8 @ 225
4 TO 5	230 X 450	M25 : Fe550 : Fe550	10	14	21.67	-23.77	39.56	1.6	0.6	4-T20 + 2-T16	T8 @ 225

Column/Wall : **C10**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
1 TO 2	230 X 450	M25 : Fe550 : Fe550	13	17	108.09	6.69	44.81	1.6	0.72	4-T20 + 2-T16	T8 @ 225
2 TO 3	230 X 450	M25 : Fe550 : Fe550	10	14	396.29	-8.73	-58.6	1.6	0.89	4-T20 + 2-T16	T8 @ 225
3 TO 4	230 X 450	M25 : Fe550 : Fe550	10	14	251.59	3.72	51.69	1.6	0.77	4-T20 + 2-T16	T8 @ 225
4 TO 5	230 X 450	M25 : Fe550 : Fe550	13	17	37.01	20.21	-32.49	1.6	0.49	4-T20 + 2-T16	T8 @ 225
5 TO 6	230 X 450	M25 : Fe550 : Fe550	9	13	17.65	41.05	-3.41	1.6	0.3	4-T20 + 2-T16	T8 @ 225

Column/Wall : **C11**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
		M25 :									

1 TO 2	230 X 530	Fe550 : Fe550	8	12	468.02	0.74	65.87	2.39	0.68	8-T20 + 2-T16	T8 @ 225
2 TO 3	230 X 530	M25 : Fe550 : Fe550	8	12	343.69	17.37	-88.17	2.39	0.93	8-T20 + 2-T16	T8 @ 225
3 TO 4	230 X 530	M25 : Fe550 : Fe550	8	12	218.33	-13.49	78.45	2.39	0.82	8-T20 + 2-T16	T8 @ 225
4 TO 5	230 X 530	M25 : Fe550 : Fe550	8	12	91.31	73.51	-54.74	2.39	0.51	8-T20 + 2-T16	T8 @ 225
5 TO 6	230 X 530	M25 : Fe550 : Fe550	1	5	32.83	54.82	6.86	2.39	0.24	8-T20 + 2-T16	T8 @ 225

Column/Wall : **C12**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
1 TO 2	230 X 450	M25 : Fe550 : Fe550	8	12	671.08	-3.75	45.83	1.6	0.75	4-T20 + 2-T16	T8 @ 225
2 TO 3	230 X 450	M25 : Fe550 : Fe550	15	19	148.35	5.29	-58.98	1.6	0.91	4-T20 + 2-T16	T8 @ 225
3 TO 4	230 X 450	M25 : Fe550 : Fe550	15	19	103.41	-6.24	51.6	1.6	0.82	4-T20 + 2-T16	T8 @ 225
4 TO 5	230 X 450	M25 : Fe550 : Fe550	15	19	49.26	16.24	39.75	1.6	0.64	4-T20 + 2-T16	T8 @ 225

Column/Wall : **C13**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
5 TO 6	230 X 300	M25 : Fe550 : Fe550	1	5	9.32	-16.91	-7.4	1.17	0.44	4-T16	T8 @ 225

Column/Wall : **C14**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
		M25 :									

5 TO 6	230 X 300	Fe550 : Fe550	1	5	0.44	-24.98	4.27	1.17	0.6	4-T16	T8 @ 225

Column/Wall : **C15**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
6 TO 7	230 X 300	M25 : Fe550 : Fe550	11	15	12.32	-19.77	-5.82	1.17	0.48	4-T16	T8 @ 225

Column/Wall : **C16**

FrameType : **Non-Ductile**

Level	Size (mm)	Material	LC	Analysis LC No	P (kN)	Mx (kNm)	My (kNm)	Pt (%)	Interaction Ratio	Main Reinforcement	Links
6 TO 7	230 X 300	M25 : Fe550 : Fe550	9	13	8.51	18.62	-7.03	1.17	0.47	4-T16	T8 @ 225